

Tax Evasion and Productivity: Identifying and Correcting for Cost Overreporting*

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Abstract

Value-added taxes (VAT) allow firms to deduct the taxes paid on purchases from the taxes collected on sales, creating an incentive for firms to overreport the purchases of inputs to reduce their tax bill. The fundamental issue in detecting input overreporting stems from two unobservables, true inputs and productivity. Because firms differ in productivity, a firm reporting high input use, for a given level of output, may be either overreporting or simply less productive. To distinguish between overreporting and productivity, I use a structural production-function approach and a benchmark group, a subset of firms that report inputs correctly. Because the benchmark firms are assumed to share the same technology as the other firms, I estimate the technology's parameters from them. Using these estimates, I compare what potentially tax-evading firms report with predictions of the true inputs they would use. I apply the method to a Colombian manufacturing survey from 1981 to 1991. During this period, the Colombian sales tax worked as a VAT and the data include a subset of firms that, due to government and market scrutiny, reported inputs correctly.

My approach yields a simple and robust statistical test to detect input overreporting. Under the null hypothesis of correct reporting, the test is agnostic about the specification of both the probability of detection and the cost of overreporting. Among the 5 largest industries, the test rejects the null hypothesis in all 4 industries subject to the sales tax, and, as expected, it fails to reject in the industry exempt from the sales tax. In these 4 industries, firms overreport on average 14.7 percent of their true inputs in the median industry. Exploiting a 1983 fiscal reform in Colombia that raised the sales tax rate, I then document that firms overreport more after the increase. The response implies a midpoint elasticity of overreporting with respect to the tax rate of 5.9.

The magnitude of detected overreporting matters for methods of estimating production functions that exploit a flexible input to identify productivity (Levinsohn and Petrin 2003; Doraszelski and Jaumandreu 2013; Akerberg et al. 2015; Gandhi et al. 2020). These methods typically assume that materials are flexible, and their demand or first-order conditions reveal productivity. However, materials are deductible under the VAT. Therefore, firms have the incentive to overreport the input these methods rely on. I show that failing to correct for

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overreporting yields larger output elasticities of materials, and more dispersed and less persistent productivity distributions. Using the production-function parameters, I estimate a structural model of input overreporting. With the model, I study how government revenue changes as expected deductions, true and fictitious, respond to the tax rate. At the baseline rate, a 1 percent increase in the sales tax rate on purchases raises expected deductions by about 1.4 percent: firms overreporting more add 0.2 points, and firms buying more materials add another 0.2. At the current rate, the government pays potentially tax-evading firms net refunds, because their claims exceed the sales tax they owe on their sales. Each 1 percent increase in the rate lowers net revenue by about 1.7 percent of that sales tax, and a cut of about 17 percent would turn the refunds into positive revenue. Undetected overreporting costs the government 4.7 percent of the sales tax firms in tax evading industries owe on their sales.

JEL Classification: H26, D22, L60, O47

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